

What the Figures Are and How to Interpret Each One

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Below is a figure-by-figure interpretation tied to the actual computed results in 3_model_comparison.json.

FIGURE 1 — pairwise_matrix.png (Divergence + Stability Heatmaps)

What it shows

A matrix view across the three checkpoints for:

- feature-importance similarity (cosine / Spearman),
- distribution divergence (JSD / Wasserstein),
- Top-K overlaps,
- stability diagnostics per output head.

Key numeric results (Feature-importance divergence)

For each pair (A, B), the audit computed:

- **Epoch 3 vs Epoch 4**

$\text{CosSim} = 0.982885, \rho_s = 0.983994, \text{JSD} = 0.071613, W = 0.002771, \text{Overlap@5} = 1.0, \text{Overlap@10} = 1.0.$

- **Epoch 3 vs Epoch 5**

$\text{CosSim} = 0.972875, \rho_s = 0.977178, \text{JSD} = 0.095514, W = 0.003945, \text{Overlap@5} = 0.8, \text{Overlap@10} = 0.9.$

- **Epoch 4 vs Epoch 5**

$\text{CosSim} = 0.997703, \rho_s = 0.996477, \text{JSD} = 0.036754, W = 0.001431, \text{Overlap@5} = 0.8, \text{Overlap@10} = 0.9.$

Interpretation

- Epoch 4 and Epoch 5 are **closest** in “what features matter” (highest CosSim/Spearman, lowest JSD/W).
- Epoch 3 is the **most different** (particularly vs Epoch 5).
- But note: “closest in attribution” is not automatically “best” (it can mean both converged to the same heuristic, including a potentially brittle one).

FIGURE 2 — `ranking_summary.png` (Tool’s Composite Score)

What it shows

A bar chart of each model’s total score (the tool produced a tie in the exported JSON).

Why it ties here

The JSON shows each model’s “total_score = 12”. This is a **known limitation** of coarse binning: when three models all satisfy thresholds similarly, a discrete scoring rubric will tie.

Scientific fix (recommended)

Replace discrete bins with a continuous scalar objective:

$$\text{Score}(M) = \alpha \text{Balance}(M) + \beta \text{Consistency}(M) + \gamma \text{Fidelity}(M) + \delta \text{Robustness}(M),$$

with continuous sub-scores defined below in Section 8.

FIGURE 3 — `model_radar.png` (Multi-Axis Quality Profile)

What it shows

Radar chart of:

- feature balance,
- interpretability (surrogate fidelity),
- attribution consistency,
- output stability proxies,
- temporal balance.

Key observed pattern

- Epoch 3 has **best feature balance** (highest entropy / lowest dominance).
- Epoch 4 has **best attribution consistency** (SHAP vs Permutation top-5 overlap = 100%).
- Epoch 5 has the **highest surrogate fidelity** (but also the most concentrated attributions).

FIGURE 4 — `divergence_metrics.png` (Pairwise Bars)

What it shows

Pairwise comparisons across CosSim, Spearman, JSD, Wasserstein, Top-K overlap.

Decision-relevant takeaways

- Epoch 4 vs Epoch 5 is the most “aligned” pair (near-identical ranking geometry).
- Epoch 3 is more “distinct,” indicating meaningful training evolution between 3 → 4/5.

FIGURE 5 — `importance_comparison.png` (Top Features Across Models)

What it shows

Side-by-side comparison of global permutation importances.

Actual top-5 permutation importances (Global)

- Epoch 3 top-5:

$[ivr, \theta]$, δ

- Epoch 4 top-5:

$[ivr, \theta]$, δ

- Epoch 5 top-5:

$[ivr, \theta]$, δ

Interpretation

- All models converge on an options-Greek + regime pressure core: $ivr, \theta, \delta, \text{pressure_down}$.
- Epoch 3/4 include **gap_risk_score**, which is directly risk-structural for overnight/event jump risk.
- Epoch 5 replaces that with **pressure_up**, suggesting a more directional/flow-tilted heuristic.

FIGURE 6–8 — `importance_heatmap_epoch_*.png` (Per-Head Importance Structure)

What they show

Heatmaps of feature importance by head.

Key per-head pattern (ROI, DTE, MAX_LOSS)

Epoch progression shows increasing dominance of ivr in ROI/DTE/MAX_LOSS heads:

- ROI head top feature ivr importance:

12.13		(E3)	
	→ 16.96		(E4)
	→ 18.75		(E5).

Interpretation

- Training is increasingly letting **IV regime** control risk-critical heads.
- This can be good (IVR is a central state variable for options), but too much dominance can reduce adaptability when IVR is noisy, stale, or regime-shifting.

FIGURE 9–11 — Decision Trees (`decision_tree_epoch_*.png`)

These are the most interpretable “mechanistic snapshots” of ROI logic.

9) Epoch 3 ROI Tree (rules excerpt)

Root split:

$$\text{ivr} \leq 0.674$$

Then a meaningful secondary structure appears involving volatility band dynamics and friction:

- splits on `bb_sigma_dyn`, δ , and `friction_ratio`.

Interpretation

- Epoch 3 expresses ROI as a function of **IV regime plus microstructure/volatility geometry** (bandwidth and friction).
- This is closer to “physics of markets”: volatility surface + flow resistance.

10) Epoch 4 ROI Tree (rules excerpt)

Root splits:

$$\text{ivr} \leq 0.687, \text{ivr} \leq 0.152, \delta \leq 0.654, \log_return \leq -1.541 \quad (\text{scaled}).$$

Interpretation

- Epoch 4 uses a cleaner “state → response” chain:
IV regime → delta structure → realized return shock gating.
- This is consistent with an options desk heuristic:
“IV regime + delta exposure + recent shock” controls ROI expectation.

11) Epoch 5 ROI Tree (rules excerpt)

Root splits tighten:

$$\text{ivr} \leq 0.496, \text{ivr} \leq 0.157, \delta \leq 0.650, \text{ivr} \leq -0.012.$$

Interpretation

- Epoch 5 becomes **more IVR-centric and self-referential** (multiple IVR thresholds).
- This increases surrogate fidelity R^2 but risks degenerating into a narrow IVR gate.

FIGURE 12–14 — Stability Plots (`stability_epoch_*_vs_*`)

These show per-head MSD, correlation, variance ratio, KS, Wasserstein.

Key ROI stability values (per pair)

- Epoch 3 vs 4 (ROI):

$$\text{MSD} = 0.011109, r = 0.01234, \text{VR} = 0.8667, D_{\text{KS}} = 0.157.$$

- Epoch 4 vs 5 (ROI):

$$\text{MSD} = 0.012646, r = -0.01799, \text{VR} = 0.9419, D_{\text{KS}} = 0.106.$$

Critical interpretation note (very important)

The correlations are near zero across many heads, while divergences of feature importance are extremely high-agreement. That combination often means **one of two things**:

1. **Sample-alignment artifact**: outputs compared across models are not perfectly aligned sample-by-sample (even if they are distributionally similar).
2. **Quasi-constant / low-signal output**: if a head's variance is low or outputs are clipped, correlation becomes unstable and uninformative.

Because your tool is meant to be a precision audit, the *scientific fix* is:

Force identical samples

That turns coherence r into a valid “same-stimulus response” measure.

FIGURE 15 — `gradient_saliency.png` (Feature + Temporal Sensitivity)

What it shows

Which features and which time regions contribute most to gradient magnitude.

Observed temporal shape (all models)

Temporal saliency is strongly front-loaded: early points in the window dominate, and saliency decays.

Interpretation

- The model behaves like a dissipative system with memory decay: recent-past segments heavily dominate the Jacobian norm.
- This is consistent with an effective **finite memory kernel** $K(t)$ that decays over the window.

A useful conceptual model:

$$\hat{y}_h \approx \int_0^1 K_h(t) \Phi_h(X(t)) dt, K_h(t) \text{ decays in } t.$$

Definitions

- $K_h(t)$: effective temporal kernel for head h .
- Φ_h : feature transform induced by latent dynamics.

FIGURE 16 — `gradient_alignment.png` (Do Models “Look” the Same Way?)

What it shows

Pairwise gradient alignment values.

Actual computed alignments

- Epoch 3 vs 4: 0.999015
- Epoch 3 vs 5: 0.999355
- Epoch 4 vs 5: 0.999489

Interpretation

- Models are nearly identical in *local sensitivity direction* in input space.
- This strongly supports “same learned geometry,” even if some outputs appear decorrelated in the stability plot (again pointing to sample-alignment/correlation-metric limitations).

FIGURE 17 — shap_comparison.png

What it shows

Mean absolute SHAP values compared across models, plus overlaps with permutation rankings.

Actual SHAP vs Permutation Top-5 overlap

Compute top-5 from permutation importances and SHAP mean absolute values:

- Epoch 3:

$$\text{Overlap} = 4/5(\theta, \text{ivr}, \delta, \text{pressure_down})$$

- Epoch 4:

$$\text{Overlap} = 5/5(\theta, \text{ivr}, \delta, \text{pressure_down}, \text{gap_risk_score})$$

- Epoch 5:

$$\text{Overlap} = 4/5(\theta, \text{ivr}, \delta, \text{pressure_down})$$

Interpretation

- Epoch 4 is the only checkpoint where two different attribution regimes (permutation disruption vs Shapley-style contribution) agree perfectly at the top.
- That is a major marker of interpretability stability.

FIGURE 18 — mutual_information.png

What it shows

Pairwise MI across features.

How to use it scientifically

If two features have high MI, they are redundant; permutation importance and SHAP can “split credit” inconsistently across redundant clusters. Therefore, the tool should also compute **cluster-aware attributions**:

$$PI(\mathcal{C}) = \sum_{j \in \mathcal{C}} PI_j,$$

where $\mathcal{C}$ is an MI-derived cluster.

This turns “best model selection” into a robust statement about which **feature families** matter, not only individual columns.

FIGURE 19–28 — `output_dist_*.png` (Per-Head Distributions)

These show histograms / KDEs (and often Q-Q style shapes) per head.

Key head statistics from JSON (absolute dispersion)

- POP standard deviation:

0.0171		(E3)	
	→ 0.00642		(E4)
	→ 0.00675		(E5).

- ROI standard deviation:

0.0723		(E3)	
	→ 0.0776		(E4)
	→ 0.0800		(E5).

Interpretation

- POP collapses in E4/E5 relative to E3 (less variability).
- ROI spreads slightly increase with training.

A physically meaningful concern: if POP becomes overly concentrated, the model’s risk perception may lose dynamic range (a sign of calibration drift or over-regularization).

FIGURE 29–31 — Fisher Comparisons (`fisher_comparison_*.png`)

These show layerwise Fisher ratio (sensitivity shift) between models.

Actual Fisher ratios (selected)

- Epoch 3 vs 4:

$\text{regime_detector} \approx 4.142, \text{expert_high} \approx 2.065, \text{expert_normal} \approx 0.206.$

- Epoch 4 vs 5:

$\text{regime_detector} \approx 1.178, \text{expert_high} \approx 0.996.$

Interpretation

- Most parameter evolution from E3 $\rightarrow$ E4 happens in **regime detection** and in the expert routing blocks.
- E4 $\rightarrow$ E5 changes are smaller, indicating **convergence** (or a move into a more rigid local basin).